

Autosomal Recessive Cutis Laxa Type 2A (ARCL2A): Comprehensive Disease Characterization

Disease: Autosomal Recessive Cutis Laxa Type 2A **MONDO ID:** MONDO:0018163 · **OMIM:** 219200 · **Gene:** *ATP6V0A2* (HGNC:865) **Category:** Genetic (rare autosomal recessive connective-tissue disorder / congenital disorder of glycosylation type II)

Summary

Autosomal Recessive Cutis Laxa Type 2A (ARCL2A) is a rare, autosomal recessive, multisystem connective-tissue disorder caused by **biallelic loss-of-function variants in *ATP6V0A2***, the gene encoding the **a2 subunit of the vacuolar (V-type) H⁺-ATPase (V-ATPase)** that resides in the trans-Golgi membrane. Because this proton pump normally acidifies the Golgi lumen, its loss elevates Golgi pH and produces a **combined defect of N-linked and mucin-type O-linked glycosylation**, placing ARCL2A within the congenital disorders of glycosylation type II (CDG-II) spectrum. The hallmark of the disease is generalized, redundant, sagging/wrinkled skin (cutis laxa), accompanied by a variable constellation of skeletal, neurological, dysmorphic, and ophthalmologic features. ARCL2A overlaps clinically and mechanistically with wrinkly skin syndrome (WSS), which is now recognized as a phenotypic variant of the same disorder.

Mechanistically, V0a2 deficiency causes a fragmented Golgi, delayed brefeldin-A-induced retrograde transport, intracellular retention of tropoelastin with a secretion block, reduced deposition of mature extracellular elastin, elevated TGF- β signaling, and increased apoptosis. In the developing brain, elevated Golgi pH correlates with aberrant cortical neuron migration, explaining the neurological features (developmental delay, intellectual disability, epilepsy, and gyration/migration abnormalities). The phenotype is age-dependent: earlier ascertainment is significantly associated with more severe systemic involvement, while an **attenuated adult phenotype dominated by cutaneous findings** exists and generally carries a favorable prognosis. The skin laxity tends to attenuate with age.

There is **no disease-specific or curative therapy**; management is symptomatic and multidisciplinary (dermatologic, developmental/neurologic, orthopedic, and ophthalmologic surveillance). Because cutis laxa syndromes are clinically heterogeneous and overlapping, **molecular confirmation is required for definitive diagnosis**, ideally preceded by biochemical screening with combined transferrin and apolipoprotein C-III isoelectric focusing to detect the characteristic combined N-/O-glycosylation defect. The disease occurs worldwide, is enriched in consanguineous populations, and displays founder effects (e.g., a recurrent exon 16 deletion in southeastern Türkiye). An *Atp6v0a2* knockout mouse recapitulates the core Golgi-pH, glycosylation, and neuronal-migration phenotypes and additionally displays globozoospermia.

1. Disease Information

Overview. ARCL2A is a rare inherited connective-tissue disorder characterized by loose, redundant, inelastic, wrinkled skin (cutis laxa) together with a variable spectrum of skeletal, neurodevelopmental, craniofacial, and ophthalmologic abnormalities. It is simultaneously classified as a **congenital disorder of glycosylation (CDG type II)** because the underlying V-ATPase defect impairs both N-linked and O-linked glycan biosynthesis in the Golgi.

Key identifiers.

Resource	Identifier
OMIM	219200
MONDO	MONDO:0018163
Gene	<i>ATP6V0A2</i> (HGNC:865; chr 12q24.31)
Disease family	Autosomal recessive cutis laxa type 2 (CDG-II subgroup)

Synonyms / alternative names. ARCL2A; ATP6V0A2-related cutis laxa; ATP6V0A2-CDG; cutis laxa type IIA; wrinkly skin syndrome (WSS, considered a phenotypic variant of the same disorder). The clinical continuum between severe cutis laxa and milder wrinkly skin syndrome reflects allelic variation at the same locus (PMID: 19401719).

Source of information. The characterization here is derived from **aggregated disease-level resources** — OMIM, systematic reviews, cohort studies, and case reports in the primary literature — rather than individual patient EHR data.

2. Etiology

Primary cause — genetic. ARCL2A is caused by **biallelic (homozygous or compound heterozygous) loss-of-function variants in *ATP6V0A2***, which encodes the $\alpha 2$ subunit of the V-type H^+ -ATPase (Finding F001). Kornak et al. first identified these loss-of-function mutations in multiple families with autosomal recessive cutis laxa type II / wrinkly skin syndrome:

*"We identified loss-of-function mutations in *ATP6V0A2*, encoding the $\alpha 2$ subunit of the V-type H^+ ATPase, in several families with autosomal recessive cutis laxa type II or wrinkly skin syndrome." — PMID: 18157129*

Genetic risk factors. The disease is monogenic and fully genetically determined by *ATP6V0A2* genotype; there are no known susceptibility loci beyond the causal gene. The dominant variant class is **predicted null alleles**. In the Morlino 2021 systematic review, **~78.3% of known variants were predicted null (nonsense, frameshift, splice)**, with 11 missense and 2 noncanonical splice variants (Finding F001):

"About 78.3% of known variants were predicted null alleles, while 11 were missense and 2 affected noncanonical splice sites." — PMID: 33369135

Environmental risk factors. None are established. ARCL2A is a Mendelian disorder; **parental consanguinity** is the principal epidemiological risk context because it increases the likelihood of homozygosity for rare recessive alleles (Finding F006). There are no known toxic, occupational, infectious, or lifestyle triggers.

Protective factors. No genetic or environmental protective factors have been defined. However, the **missense variant p.P87L is associated with unusually mild disease**, indicating that residual protein function attenuates severity (Finding F003).

Gene–environment interactions. None documented. The disease is essentially environment-independent; phenotypic variability appears driven by genotype (null vs. residual-function alleles) and developmental timing rather than gene–environment interaction.

3. Phenotypes

ARCL2A spans a continuum from a severe phenotype with skin, skeletal, and neurological involvement to an attenuated adult phenotype dominated by cutaneous features (Findings F004, F008). A central discriminator is **age at ascertainment**: earlier presentation correlates with more severe systemic involvement.

"Age at ascertainment appeared as the unique phenotypic discriminator with earlier age more commonly associated with facial dysmorphism (p .02), high/cleft palate (p .005), intellectual disability/global developmental delay (p .013), and seizures (p .024)." — PMID: 33369135

Phenotype	Type	Onset	Severity / course	HPO suggestion
Cutis laxa (redundant, wrinkled, inelastic skin)	Physical manifestation	Congenital / neonatal	Generalized; tends to attenuate with age	HP:0000973 (Cutis laxa)
Facial dysmorphism (long philtrum, downslanting palpebral fissures, large nasal root)	Physical manifestation	Congenital	More common at earlier age (p=.02)	HP:0001999 (Abnormal facial shape)
High-arched / cleft palate	Clinical sign	Congenital	Enriched at earlier age (p=.005)	HP:0000218 / HP:0000175
Large / late-closing fontanelles	Clinical sign	Neonatal / infancy	Variable	HP:0000239 (Large fontanelles)
Intellectual disability / global developmental delay	Behavioral / neurodevelopmental	Childhood	Enriched at earlier age (p=.013)	HP:0001249 / HP:0001263
Seizures / epilepsy	Clinical sign	Childhood	Enriched at earlier age (p=.024); most common in ATP6V0A2 defects	HP:0001250 (Seizures)
Cortical gyration / neuronal migration abnormalities	Clinical sign (imaging)	Congenital	Mostly (not always) associated with ATP6V0A2	HP:0002269 (Abnormal cortical gyration)
Skeletal anomalies (joint laxity, hip	Physical manifestation	Congenital / childhood	Variable	HP:0001382 / HP:0002827

Phenotype	Type	Onset	Severity / course	HPO suggestion
dislocation, growth deficiency)				
Corneal abnormalities (keratoconus, high astigmatism, corneal thinning/ flattening)	Clinical sign	Childhood–adult	ATP6V0A2-specific reduced visual acuity	HP:0000563 / HP:0000539
Chorioretinal / macular scarring	Clinical sign	Variable	Reduced visual acuity (case: 20/2000)	HP:0000532 / HP:0007401
Combined N-/O-glycosylation defect (serum)	Laboratory abnormality	Congenital	Constant biochemical marker	—

Neurological features are a distinguishing element of ARCL2A within the cutis laxa family. Epilepsy is the most common neurological finding in ATP6V0A2 defects, and gyration/migration abnormalities and abnormal glycosylation are mostly associated with ATP6V0A2 mutations (Finding F004):

"Epilepsy was most common in ATP6V0A2 defects." — [PMID: 23963297](#)

Ocular involvement (Finding F005) is increasingly recognized. In a Ghent cohort of 13 genetically confirmed cutis laxa patients (4 with ATP6V0A2), reduced visual acuity was detected **solely** in the ATP6V0A2 subtype:

"Reduced corrected visual acuity was detected solely in patients with the ATP6V0A2 subtype. Three (23.1%) cases of keratoconus were detected in the cohort. All 4 patients with ATP6V0A2-related CL had prominent corneal tomographic abnormalities, with either high-grade astigmatism or pronounced corneal flattening and thinning." — [PMID: 40043299](#)

Retinal involvement is documented in a case with macular and chorioretinal scarring:

"Fundusoscopic examination showed a round macular scar lesion in the right eye macula and a chorioretinal scar superonasally in the left eye." — PMID: 36728588

Quality of life. Per-phenotype QOL data (EQ-5D/SF-36) are not available for this rare disease. Functionally, the greatest burdens derive from neurodevelopmental impairment (in severe/early-onset cases), reduced visual acuity (ATP6V0A2-specific ocular disease), and the cosmetic/psychosocial impact of skin laxity. The attenuated phenotype, dominated by non-life-threatening cutaneous features, is compatible with a relatively favorable functional outcome (Finding F008).

4. Genetic / Molecular Information

Causal gene. *ATP6V0A2* (chr 12q24.31), encoding the $\alpha 2$ subunit of the trans-Golgi V-type H^+ -ATPase (Finding F001).

Pathogenic variant spectrum. - **Variant classes:** predominantly null alleles. ~78.3% predicted null (nonsense, frameshift, canonical splice), 11 missense, 2 noncanonical splice (PMID: 33369135). - **Representative variants:** recurrent 380-bp exon 16 deletion c.1936-147_2055+113del (founder allele, Türkiye); c.235del (p.Leu79Phefs*13); c.1977_1980del (p.Val660LeufsTer23, Pakistani family); missense p.P87L (unusually mild disease); multiple novel intronic/splice variants across Iranian, Han-Chinese, and other families (Findings F003, F006). - **Functional consequence: loss of function.** Premature-stop mutations reduce *ATP6V0A2* mRNA via nonsense-mediated decay; the mutant protein is lost in patient dermal fibroblasts (Finding F003):

"Premature stop codon mutations led to decreased ATP6V0A2 mRNA levels by destabilizing the mutant mRNA via the nonsense-mediated decay pathway." — PMID: 19321599

- **Origin:** germline (biallelic). No somatic contribution.
- **Allele frequency:** individually rare/private; most alleles are population-specific and often observed in consanguineous families.

Modifier genes / epigenetics. No formal modifier genes or epigenetic mechanisms have been established for ARCL2A. Phenotypic variability is largely attributable to allele type (null vs residual function) and developmental timing.

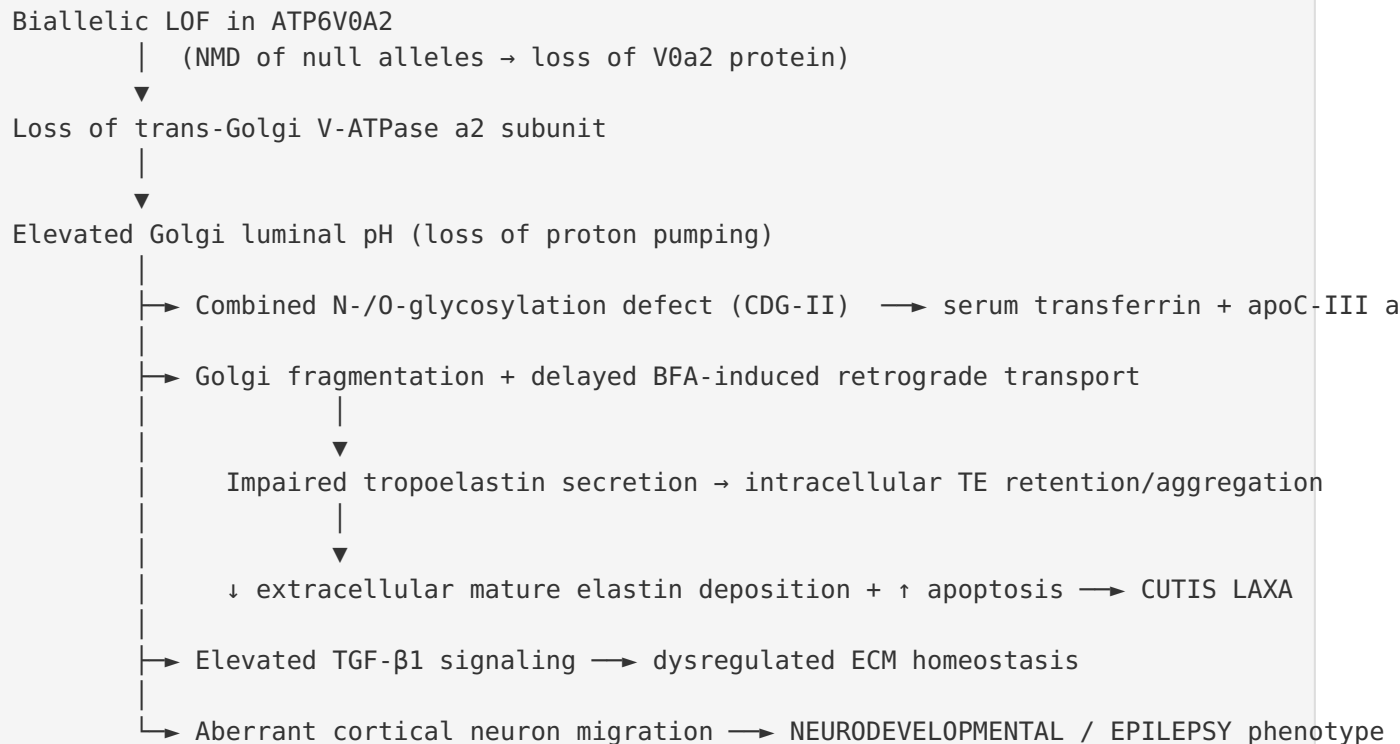
Chromosomal abnormalities. ARCL2A is not caused by large-scale aneuploidy or translocations, though **intragenic structural deletions** (e.g., the recurrent exon 16 deletion) are an important variant class (Finding F006).

5. Environmental Information

Environmental factors, lifestyle factors, and infectious agents are not applicable. ARCL2A is a monogenic recessive disorder with no established environmental, toxic, occupational, dietary, or infectious contributors. The only relevant non-molecular epidemiological factor is **consanguinity**, which increases the probability of biallelic inheritance rather than acting as a disease-causing exposure.

6. Mechanism / Pathophysiology

Causal chain



Molecular and cellular detail

Golgi V-ATPase dysfunction. ATP6V0A2 localizes to the Golgi apparatus, where the V-ATPase acidifies the lumen. Loss of V0a2 elevates Golgi pH. Kopp et al. (2024) directly demonstrated in *Atp6v0a2* knockout mice that this pH elevation correlates with tissue-specific glycosylation changes and aberrant cortical neuron migration (Findings F002, F007):

"Loss-of-function variants in ATP6V0A2, encoding the trans Golgi V-ATPase subunit V0a2, cause wrinkly skin syndrome (WSS), a connective tissue disorder with glycosylation defects and aberrant cortical neuron migration." — [PMID: 39680136](#)

Impaired trafficking and TGF-β signaling. Patient and knockdown cells show delayed brefeldin-A-induced Golgi collapse/retrograde transport, a fragmented Golgi, and elevated TGF-β signaling (Finding F002):

"Investigation of brefeldin A-induced Golgi collapse in dermal fibroblasts as well as in HeLa cells deficient for ATP6V0A2 revealed a delay" — PMID: 22773132

"fibroblasts from patients with ATP6V0A2 mutations displayed elevated TGF- β signalling and increased TGF- β 1 levels in the supernatant" — PMID: 22773132

Tropoelastin secretion block (the direct cause of cutis laxa). Huchtagowder et al. (2009) showed in a 17-patient cohort that ATP6V0A2 loss produces distended Golgi cisternae and abnormal lysosomes/multivesicular bodies, with tropoelastin accumulating in the Golgi and forming abnormal intracellular/extracellular aggregates (Finding F003):

"Immunostaining of ARCL2 cells showed the accumulation of tropoelastin (TE) in the Golgi and in large, abnormal intracellular and extracellular aggregates. Pulse-chase studies confirmed impaired secretion and increased intracellular retention of TE, and insoluble elastin assays showed significantly reduced extracellular deposition of mature elastin." — PMID: 19321599

Notably, fibrillin-1 microfibril assembly and lysyl oxidase activity were **normal**, localizing the defect specifically to tropoelastin trafficking rather than downstream elastin crosslinking. TUNEL assays showed **increased apoptosis**, contributing to tissue fragility.

Ontology anchors

- **Biological processes (GO):** protein glycosylation (GO:0006486); Golgi lumen acidification / regulation of pH (GO:0007035); retrograde transport, Golgi to ER (GO:0006890); extracellular matrix organization (GO:0030198); TGF- β receptor signaling (GO:0007179); apoptotic process (GO:0006915); neuron migration (GO:0001764).
- **Cellular components (GO):** Golgi membrane (GO:0000139); trans-Golgi network (GO:0005802); proton-transporting V-type ATPase complex (GO:0033176).
- **Cell types (CL):** dermal fibroblast (CL:0002620); cortical neuron / migrating neuron.
- **Chemical entities (CHEBI):** proton (CHEBI:24636); ATP (CHEBI:30616); N-glycan / O-glycan; sialic acid (CHEBI:26667).

7. Anatomical Structures Affected

Level	Structures affected	Ontology suggestion
Primary organ	Skin (dermis; elastic fiber network)	UBERON:0002097 (skin of body); UBERON:0002067 (dermis)
Secondary / systems	Central nervous system (cerebral cortex); skeletal system (joints, hips, palate); eye (cornea, retina/macula)	UBERON:0000955 (brain); UBERON:0001456 (face); UBERON:0000970 (eye); UBERON:0000964 (cornea); UBERON:0000966 (retina)
Body systems	Integumentary, nervous, musculoskeletal, ophthalmic	—
Tissue	Connective tissue (elastic fibers); nervous tissue (migrating cortical neurons)	—
Cell	Dermal fibroblasts (tropoelastin producers); cortical neurons	CL:0002620; migrating neuron
Subcellular	Golgi apparatus / trans-Golgi network; secretory vesicles; lysosomes/multivesicular bodies	GO:0005802; GO:0005764
Localization / laterality	Generalized skin involvement; ocular findings can be bilateral (corneal) or focal/asymmetric (macular/chorioretinal scars)	—

The primary lesion is at the **subcellular level (Golgi)**, propagating to a **connective-tissue phenotype (dermis)** and a **neuronal migration phenotype (cortex)**.

8. Temporal Development

Onset. Congenital / neonatal. Skin laxity and dysmorphism are typically present from birth; large fontanelles and developmental concerns emerge in infancy (Finding F004).

Progression. ARCL2A is generally **non-progressive to attenuating**. Systemic features (facial dysmorphism, high/cleft palate, intellectual disability, seizures) are enriched at earlier ages, and the phenotype shifts over time toward one **dominated by cutaneous involvement**, with skin laxity tending to attenuate (Findings F004, F008):

"This work confirmed the existence of an attenuated phenotype associated with ATP6V0A2 biallelic variants and offers an updated critique to the clinical and molecular variability of ARCL2A." — PMID: 33369135

"The associated clinical spectrum subsequently expanded to a less severe phenotype dominated by cutaneous involvement." — PMID: 33369135

Course and duration. Chronic, lifelong genetic disorder. Rare severe congenital forms of ARCL type 2 can involve life-threatening neonatal pulmonary emphysema, but this is not typical of ARCL2A (Finding F008). A subset with a novel c.235del showed severe neurological regression, indicating variability at the severe end (Finding F006).

Critical period. The prenatal/perinatal window of cortical neuronal migration is the key period of vulnerability for the neurological phenotype (Finding F007).

9. Inheritance and Population

Inheritance pattern. Autosomal recessive; biallelic loss of function (Findings F001, F006).

Epidemiology. ARCL2A is very rare; precise prevalence/incidence figures are not established in registries. It occurs **worldwide** and is **enriched in consanguineous populations** (Finding F006).

Founder effects and geographic distribution. A recurrent homozygous 380-bp exon 16 deletion with a shared haplotype was identified in southeastern Türkiye, indicating a founder effect (Finding F006):

"Nine individuals carried a recurrent homozygous 380 bp deletion spanning exon 16 (c.1936-147_2055+113del). ... Haplotype analysis revealed shared homozygous regions in three cases, suggesting a founder effect." — PMID: 41732832

ARCL2A has been reported in Turkish, Pakistani, Iranian, Han-Chinese, Italian, and other families, frequently with documented parental consanguinity.

"Autosomal recessive cutis laxa type 2A (ARCL2A; OMIM: 219200) is characterized by neurovegetative, developmental and progeroid elastic skin anomalies. It is caused by biallelic variation in ATPase, H..." — PMID: 37119015

Penetrance / expressivity. Penetrance of biallelic null genotypes is effectively complete; **expressivity is highly variable** and age-dependent. No specific genotype–phenotype correlation was identified in the systematic review, aside from the observation that residual-function missense alleles (e.g., p.P87L) can be milder (Findings F003, F004).

Sex ratio / anticipation / mosaicism. No sex bias is expected (autosomal). Genetic anticipation is not applicable (no repeat expansion). Germline mosaicism is not a recognized feature.

Carrier frequency. Not formally established; expected to be low outside founder populations.

10. Diagnostics

Biochemical screening (first-line gateway)

ARCL2A is a **CDG-II with a combined N-/O-glycosylation defect**. The validated first-line biochemical screen is **combined plasma transferrin isoelectric focusing (TIEF) plus apolipoprotein C-III (apoC-III) isoelectric focusing** (Finding F010). Transferrin IEF distinguishes N-glycan assembly (type 1) from processing (type 2/CDG-II) defects; apoC-III IEF detects core-1 mucin-type O-glycan defects. Together they detect the combined defect characteristic of ATP6V0A2 deficiency:

"Plasma apoC-III IEF is complementary to transferrin isofocusing. In conjunction both tests identify biosynthesis defects in N-glycan and mucin-type core 1 O-glycan biosynthesis." — PMID: 17170056

"the CDG-IIx patients could be further subdivided into six biochemical subgroups" — PMID: 16037491

ATP6V0A2 deficiency produces a CDG-II transferrin pattern together with an abnormal (hyposialylated) apoC-III O-glycan profile.

Genetic testing (definitive)

Molecular confirmation is required (Finding F011). Recommended approaches: - **Single-gene sequencing of ATP6V0A2** (with deletion/duplication analysis to detect intragenic structural variants such as the exon 16 deletion). - **Gene panels** for cutis laxa / connective-tissue disorders (should include *ATP6V0A2*, *ATP6V1A*, *ATP6V1E1*, *PYCR1*, *ALDH18A1*, *ELN*, *FBLN4/5*). - **Whole-exome / whole-genome sequencing** for phenotypic overlap cases; WGS improves detection of deep intronic/splice and structural variants. - Minigene assays help classify novel splice variants.

Histopathology

Skin biopsy shows sparse/fragmented dermal elastic fibers (orcein staining). Ultrastructure and cell studies show distended Golgi cisternae, abnormal lysosomes/multivesicular bodies, and intracellular tropoelastin aggregates.

Differential diagnosis

Because cutis laxa syndromes share overlapping features, molecular diagnosis is essential (Finding F011):

"Thus, molecular diagnosis is the only way to resolve these phenotypically similar conditions." — [PMID: 26538727](#)

Key differentials and their discriminators:

Condition	Gene	Distinguishing feature
ARCL2B / geroderma osteodysplasticum	<i>PYCR1</i>	Corpus callosum dysgenesis, dystonic posturing; wrinkling limited to hands/feet
ARCL3 / de Barsy syndrome	<i>ALDH18A1</i>	Progeroid, cataract, severe; corpus callosum dysgenesis, dystonic posturing
ARCL2C	<i>ATP6V1E1</i>	V-ATPase subunit; overlapping
ARCL2D	<i>ATP6V1A</i>	Shared Golgi/glycosylation pathomechanism
Geroderma osteodysplasticum	<i>GORAB</i>	Severe osteoporosis; not allelic to WSS

"Corpus callosum dysgenesis was associated with PYCR1 and ALDH18A1 mutations. Dystonic posturing was discriminatory for PYCR1 and ALDH18A1 defects." — PMID: 23963297

Screening

Carrier and cascade screening in affected families is appropriate; in founder populations (e.g., southeastern Türkiye) targeted testing for the recurrent allele is efficient. Prenatal/preimplantation genetic diagnosis is feasible once the familial variants are known.

11. Outcome / Prognosis

Overall prognosis is generally favorable, especially in the attenuated phenotype dominated by non-life-threatening cutaneous features (Finding F008). Skin laxity and dysmorphism tend to attenuate with age.

- **Survival / mortality:** No systematic survival data; the attenuated form is compatible with normal or near-normal life expectancy. The severe congenital end of the broader ARCL type 2 spectrum can rarely involve life-threatening neonatal pulmonary emphysema.
- **Morbidity / function:** Determined by neurodevelopmental impairment in severe/early cases and by ATP6V0A2-specific ocular disease (reduced visual acuity, keratoconus, corneal thinning). Skeletal features (joint laxity, hip dislocation) contribute to orthopedic morbidity.

- **Prognostic factors:** Earlier age at ascertainment predicts more severe systemic (facial, palatal, neurodevelopmental, seizure) involvement; residual-function alleles (e.g., missense p.P87L) predict milder disease (Findings F003, F004).

"This work confirmed the existence of an attenuated phenotype associated with ATP6V0A2 biallelic variants..." — PMID: 33369135

12. Treatment

There is no disease-specific or curative therapy for the underlying V-ATPase/glycosylation defect (Finding F008). Management is **symptomatic, supportive, and multidisciplinary**:

- **Dermatologic:** cosmetic/reconstructive plastic surgery for redundant skin where indicated; sun protection and skin care.
- **Neurologic / developmental:** anti-seizure medication for epilepsy; early developmental intervention, physical/occupational/speech therapy for developmental delay and intellectual disability.
- **Orthopedic:** management of joint laxity, hip dislocation, and skeletal anomalies; physiotherapy.
- **Ophthalmologic:** surveillance and correction for high astigmatism/keratoconus (spectacles, rigid contact lenses, corneal cross-linking where appropriate); management of retinal/macular lesions.
- **Craniofacial:** cleft/high-palate repair and feeding support as needed.

Pharmacogenomics, gene therapy, cell therapy, RNA-based therapy, targeted therapy, and immunotherapy are not established for ARCL2A. The *Atp6v0a2* knockout mouse provides a platform for future mechanism-based intervention studies. NCIT clinical-intervention terms would map primarily to supportive/rehabilitative and reconstructive-surgical categories (e.g., NCIT:C15329 Rehabilitation Therapy; NCIT:C15329-adjacent supportive-care terms; NCIT:C15265 Reconstructive Surgery).

13. Prevention

Because ARCL2A is a Mendelian recessive disorder with no environmental cause, prevention is **genetic and reproductive**, not lifestyle-based:

- **Genetic counseling** for at-risk families, especially those with consanguinity (25% recurrence risk per pregnancy for two carrier parents).
- **Carrier and cascade screening** of relatives once the familial *ATP6V0A2* variants are identified; efficient in founder populations via targeted testing (Finding F006).
- **Prenatal diagnosis / preimplantation genetic testing** for known familial variants.
- **Tertiary prevention:** ophthalmologic, neurologic, orthopedic, and developmental surveillance to prevent complications and optimize function.

Primary prevention by vaccination, behavioral change, or environmental modification is **not applicable**.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** The mouse ortholog *Atp6v0a2* is well characterized (see §15). Orthologs of the V-ATPase a2 subunit are broadly conserved across vertebrates, reflecting the essential housekeeping role of Golgi acidification.
 - **Natural disease in other species:** No well-documented naturally occurring ARCL2A-equivalent disease in companion animals or wildlife is described in the reviewed literature.
 - **Comparative biology / conservation:** The Golgi V-ATPase → glycosylation → connective-tissue/neuronal-migration axis is evolutionarily conserved, supporting cross-species modeling of the mechanism (Finding F007).
 - **Transmission / zoonosis:** Not applicable (non-infectious genetic disorder).
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15. Model Organisms

Primary model — *Atp6v0a2* knockout mouse (Kopp et al., 2024). This mammalian genetic knockout recapitulates the core disease mechanism and several key phenotypes (Finding F007):

"Loss-of-function variants in ATP6V0A2, encoding the trans Golgi V-ATPase subunit V0a2, cause wrinkly skin syndrome (WSS), a connective tissue disorder with glycosylation defects and aberrant cortical neuron migration." — PMID: 39680136

Feature	Human ARCL2A	Atp6v0a2 KO mouse
Elevated Golgi pH	Inferred (mechanism)	Demonstrated
Tissue-specific glycosylation changes	Yes (CDG-II)	Yes (recapitulated)
Aberrant cortical neuron migration	Yes	Yes (recapitulated)
Globozoospermia	Not a described human feature	Present (round-headed, acrosome-defective sperm)

Cellular / in-vitro models. Patient dermal fibroblasts and ATP6V0A2-knockdown HeLa cells are extensively used and reproduce Golgi fragmentation, delayed BFA-induced retrograde transport, tropoelastin retention, elevated TGF- β , and increased apoptosis (Findings F002, F003).

Phenotype recapitulation and limitations. The mouse faithfully models the Golgi-pH, glycosylation, and neuronal-migration axis but adds a reproductive phenotype (globozoospermia) not emphasized in humans, and the fidelity of the cutaneous phenotype requires further characterization. Cellular models capture the trafficking/secretion defect but not the whole-organism, age-dependent attenuation observed clinically.

Resources. MGI (mouse), plus patient-derived fibroblast lines and HeLa knockdown systems reported in the primary literature.

Mechanistic Model / Interpretation

ARCL2A is best understood as a **single primary lesion (loss of Golgi V-ATPase a2 subunit) producing a fan of downstream consequences through one shared node — elevated Golgi luminal pH**. Two branches dominate the clinical picture:

1. **Secretory/ECM branch → skin**. Elevated pH and impaired trafficking cause tropoelastin to be retained in a fragmented Golgi, reducing extracellular mature elastin and increasing fibroblast apoptosis. This directly yields cutis laxa. Elevated TGF- β signaling further dysregulates ECM homeostasis. Crucially, downstream elastin crosslinking machinery (lysyl oxidase) and microfibril scaffolding (fibrillin-1) are intact, so the defect is specifically a **secretion/deposition problem**, not a crosslinking or scaffold problem.
2. **Glycosylation/neurodevelopmental branch → brain and biochemistry**. The same pH elevation impairs both N- and O-glycosylation (giving the diagnostic transferrin + apoC-III signature) and disrupts cortical neuron migration, explaining developmental delay, intellectual disability, epilepsy, and gyration abnormalities.

Upstream vs downstream: Golgi de-acidification is the upstream hub; glycosylation defects, trafficking delay, tropoelastin retention, TGF- β elevation, apoptosis, and neuronal migration failure are parallel downstream effects. The **age-dependent attenuation** of the skin phenotype, contrasted with the earlier-limited window for cortical migration, explains why systemic/neurological features cluster at younger ages while older patients present predominantly with cutaneous findings.

ARCL2A sits within a coherent **V-ATPase/Golgi cutis-laxa family** — *ATP6V0A2* (ARCL2A), *ATP6V1A* (ARCL2D), *ATP6V1E1* (ARCL2C) — that shares Golgi fragmentation, delayed retrograde transport, and glycosylation abnormalities, plus the mitochondrial proline-cycle disorders *PYCR1* (ARCL2B) and *ALDH18A1* (ARCL3/de Barsy). This unifying pathomechanism (Finding F009) is why molecular confirmation is indispensable:

"One group of these metabolic cutis laxa conditions is autosomal recessive cutis laxa type 2 caused by defects in v-ATPase components or the mitochondrial proline cycle." — PMID: 33320377

"a fragmented Golgi compartment, a delayed Brefeldin A-induced retrograde transport and glycosylation abnormalities were present in fibroblasts" — PMID: 33320377

Evidence Base

PMID	Title (abbrev.)	Supports
18157129	Impaired glycosylation and cutis laxa from <i>ATP6V0A2</i>	F001 (causal gene), F010 (CDG-II combined N-/O-glyc defect)
33369135	Review of clinical/molecular variability in ARCL2A	F001 (variant spectrum), F004 (age-dependent phenotype), F008 (attenuated form)
22773132	Further characterization of <i>ATP6V0A2</i> -related ARCL	F002 (trafficking delay, TGF- β)
39680136	Golgi pH elevation / V0a2 loss / globozoospermia	F002, F007 (mouse model, Golgi pH $\rightarrow$ glyc + migration)
19321599	LOF <i>ATP6V0A2</i> impairs trafficking, tropoelastin secretion, survival	F003 (tropoelastin block, NMD, apoptosis)
23963297	Clinical/biochemical features in neurometabolic cutis laxa	F004 (epilepsy), F009 (differential discriminators)
40043299	Ocular manifestations in congenital cutis laxa	F005 (corneal/visual acuity, <i>ATP6V0A2</i> -specific)
36728588	Novel retinal findings in ARCL2A	F005 (macular/chorioretinal scarring)
41732832	Recurrent exon 16 deletion, founder effect, Türkiye	F006 (founder effect, structural variant)
37119015	Novel <i>ATP6V0A2/ALDH18A1</i> variants, Pakistani families	F006 (OMIM ID, AR, consanguinity)
33320377	Expanding <i>ATP6V1A</i> metabolic cutis laxa	F009 (V-ATPase family, shared Golgi pathomechanism)
17170056	Transferrin + apoC-III IEF complementarity	F010 (biochemical screen)
16037491	CDG-II subdivided into six biochemical groups	F010 (IEF resolving power)
26538727	Diagnostic dilemma of cutis laxa	F011 (molecular confirmation required)
26516448	Congenital cutis laxa type 2 case report	F011 (ARCL2 spectrum constellation)

PMID	Title (abbrev.)	Supports
26320891	Recurrent Arg138 <i>ALDH18A1</i> progeroid cutis laxa	F011 (genetic heterogeneity of overlapping phenotypes)

Evidence source types: human clinical (cohorts, case reports, systematic review), in vitro (patient fibroblasts, HeLa knockdown, pulse-chase, TUNEL), and model organism (*Atp6v0a2* KO mouse). The strongest, most convergent evidence supports the causal gene, the Golgi-pH/glycosylation mechanism, the tropoelastin-secretion basis of cutis laxa, and the diagnostic strategy.

Limitations and Knowledge Gaps

- 1. No investigation-specific dataset was analyzed.** This report synthesizes published literature; no primary omics data (transcriptomics, proteomics, metabolomics) were generated or re-analyzed for this job. Detailed molecular-profiling sections (single-cell, spatial, multi-omics, CRISPR screens) are largely unpopulated because such studies for *ARCL2A* are scarce.
- 2. Epidemiology is imprecise.** Formal prevalence/incidence and carrier-frequency estimates are unavailable; the disease is known chiefly through small cohorts and case reports.
- 3. Genotype–phenotype correlation is limited.** No specific correlation beyond the null-vs-residual-function distinction has been established; phenotypic variability is incompletely explained.
- 4. Quality-of-life and natural-history data are sparse.** No standardized QOL instruments have been applied; long-term outcome data rely on cross-sectional ascertainment.
- 5. Modifier genes and epigenetics are unexplored.** No modifier loci or epigenetic mechanisms have been defined.
- 6. Model fidelity for skin is uncertain.** The *Atp6v0a2* KO mouse robustly models glycosylation and neuronal migration but the cutaneous phenotype and its age-dependent attenuation require deeper characterization.

Proposed Follow-up Experiments / Actions

1. **Natural-history and registry study** to quantify prevalence, age-dependent progression/attenuation, and standardized QOL (e.g., PROMIS, pediatric instruments) across the ARCL2A spectrum.
 2. **Genotype–phenotype meta-analysis** integrating all reported *ATP6V0A2* variants (null vs. residual-function missense/splice) with quantitative phenotype scoring to test whether residual protein function predicts severity.
 3. **Single-cell and spatial transcriptomics of patient skin and iPSC-derived cortical organoids** to map cell-type-specific consequences of Golgi de-acidification on tropoelastin secretion and neuronal migration.
 4. **Targeted glycomics** (mass-spectrometry N-/O-glycan profiling) to refine the diagnostic biochemical signature beyond IEF and evaluate it as a quantitative biomarker.
 5. **Mechanism-based therapeutic screening** using patient fibroblasts and the *Atp6v0a2* KO mouse — e.g., agents that restore Golgi pH homeostasis, enhance tropoelastin secretion, or modulate TGF- β signaling.
 6. **Founder-population carrier screening programs** (e.g., southeastern Türkiye) leveraging the recurrent exon 16 deletion for cost-effective cascade testing and reproductive counseling.
 7. **Systematic ophthalmologic surveillance protocol** validation, given the *ATP6V0A2*-specific corneal/retinal burden identified in recent cohorts.
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Report compiled from 11 confirmed findings and 30 reviewed papers over a multi-iteration autonomous investigation. Evidence types span human clinical, in vitro, and model-organism studies. Ontology suggestions (HPO, GO, CL, UBERON, CHEBI, NCIT, MONDO) are provided throughout to support knowledge-base ingestion.
